

Deploy IBM Bob with IBM Planning Analytics SaaS

This guide provides step-by-step instructions for connecting IBM Bob to an IBM Planning Analytics MCP server.

Overview

IBM Bob can connect to IBM Planning Analytics through the Model Context Protocol (MCP), enabling AI-powered interactions with your Planning Analytics data, cubes, and TM1 servers.

(Read more about IBM Planning Analytics MCP Server [here.](#))

Prerequisites

Before you begin, ensure you have:

1. **IBM Bob** installed and running
2. **IBM Planning Analytics** instance (Planning Analytics as a Service in this documentation) with **Planning Analytics Agent** entitlement
3. **Authentication credentials** for your Planning Analytics instance
4. **Network access** to your Planning Analytics server

Connection Endpoints

IBM Planning Analytics MCP tools are available for Planning Analytics as a Service (since 1/28/2026)

SaaS

- Uses HTTPS protocol
- Planning Analytics organizes MCP tools into functional groupings based on the capabilities that they provide.
 - TM1 cube tools
 - Explore cube structures, dimensions, members, and data with natural-language or structured queries.
 - `https://{PAW_HOST}/api/{PAW_TENANT_ID}/v0/agent-ai/cube/mcp`

- TM1 server tools
 - This tool set contains server tools for managing and interacting with TM1-Æ processes and servers. Each tool includes input arguments, outputs, and example usage in natural language.
 - `https://{PAW_HOST}/api/{PAW_TENANT_ID}/v0/agentica-ai/analysis/mcp`

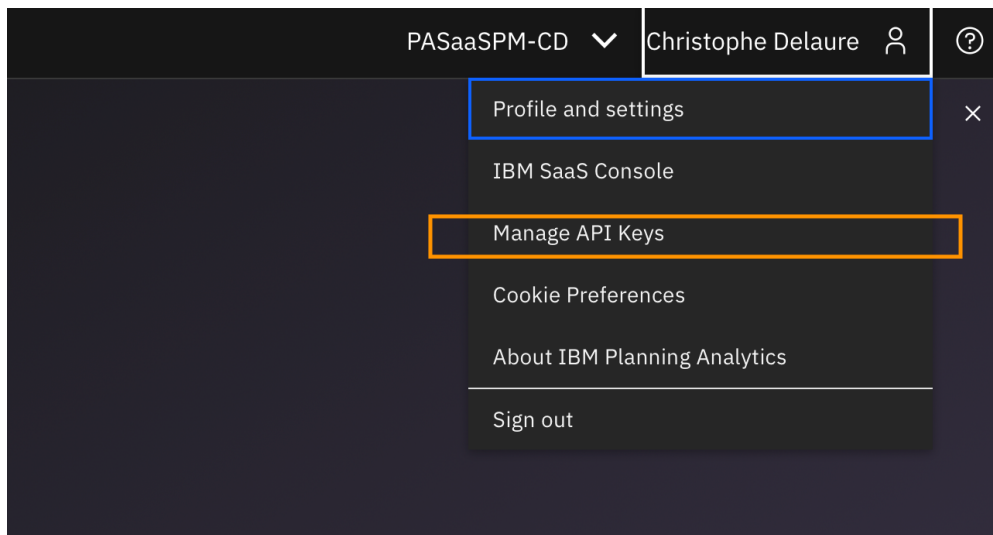
Note: the TENANT_ID can be found in the URL for workspace as the 2nd ID

Connection Instructions

Step 1: Get the API Key for IBM Planning Analytics

For SaaS Deployment:

- **URL:** Your PA SaaS region (e.g., us-east-1)
- **Tenant ID:** Your unique database identifier which is available in the url `https://us-east-1.planninganalytics.saas.ibm.com/?accountId=XXXXXXXX&tenantId=XXXXXR&perspective=pa-home`
- **API Key:** Your PA SaaS API key from you User Menu



Step 2: Create Base64 Encoded Credentials

You need to encode your credentials in Base64 format for the Authorization header.

Format – `apikey:xxxxxxxxx`

è Just Ask Bob to do the Base64 encoding

Step 3: Edit the IBM Bob MCP Settings File

1. Go into Setting (top right gear icon)
2. Go into the MCP configuration panel
3. Edit the global MCP file or the project MCP file
 - a. The Global MCP file will make the Tools available to any interaction
4. Copy and paste the file changing the URLs, API Token
5. Save the configuration
6. Close IBM Bob
7. Start IBM Bob

Configuration file example

```
{
  "mcpServers": {
    "your-server-name": {
      "type": "streamable-http",
      "url": "YOUR_PA_MCP_ENDPOINT",
      "headers": {
        "Authorization": "Basic YOUR_BASE64_CREDENTIALS"
      },
      "timeout": 300,
      "alwaysAllow": [
        "get_cubes_that_may_answer_query",
        "get_available_tm1_servers",
        "get_data_from_data_explorer",
        "lookup_potential_members",
        "get_cube_sample_members"
      ],
      "disabled": false
    }
  }
}
```

You will need to find you TENANT_ID and you API Key

- The TENANT ID can can be found in your Planning Analytics Workspace URL
https://us-east-1.planninganalytics.saas.ibm.com/api/{TENANT_ID}/v0/agentica/cube/mcp
- The API key should be encoded in 64Bits per step above

Example: Planning Analytics as a Service (PAaaS) Server

```
{
  "mcpServers": {
    "my-pa-saas-server": {
      "type": "streamable-http",
      "url": "https://us-east-1.planninganalytics.saas.ibm.com/api/{TENANT_ID}/v0/agenticaicube/mcp",
      "headers": {
        "Authorization": "Basic XXXXXXXXXXXXX"
      },
      "timeout": 300,
      "alwaysAllow": [
        "get_cubes_that_may_answer_query",
        "get_available_tm1_servers",
        "get_data_from_data_explorer",
        "lookup_potential_members",
        "get_cube_sample_members"
      ],
      "disabled": false
    }
  }
}
```

Example 2: Multiple Servers

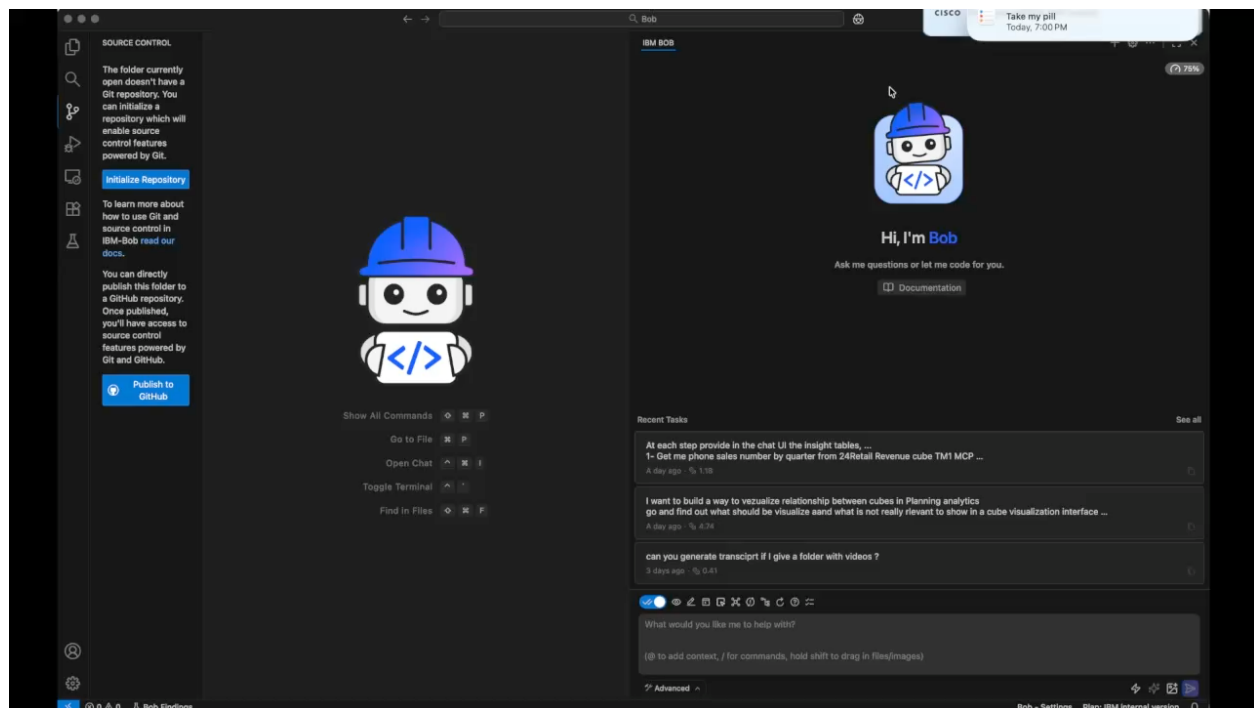
You can configure multiple PA servers:

```
{
  "mcpServers": {
    "pa-production": {
      "type": "streamable-http",
      "url": "http://pa-prod.company.com/api/v0/agenticaicube/mcp",
      "headers": {
        "Authorization": "Basic XXXXXXXXX="
      },
      "timeout": 300,
      "alwaysAllow": [
        "get_cubes_that_may_answer_query",
        "get_available_tm1_servers",
        "get_data_from_data_explorer",
        "lookup_potential_members",
        "get_cube_sample_members"
      ],
      "disabled": false
    },
    "pa-development": {
```

```

"type": "streamable-http",
"url": "http://pa-dev.company.com/api/v0/agent-ai/cube/mcp",
"headers": {
  "Authorization": "Basic XXXXXXXX="
},
"timeout": 300,
"alwaysAllow": [
  "get_cubes_that_may_answer_query",
  "get_available_tm1_servers",
  "get_data_from_data_explorer",
  "lookup_potential_members",
  "get_cube_sample_members"
],
"disabled": false
}
}
}

```



Step 4: Configuration Parameters Explained

Parameter	Description	Required	Default
type	Connection type. Must be "streamable-http"	Yes	-
url	Full MCP endpoint URL	Yes	-
headers.Authorization	Base64 encoded credentials with "Basic " prefix	Yes	-
timeout	Request timeout in seconds	No	300
alwaysAllow	Array of tool names that don't require user approval	No	[]
disabled	Set to true to disable the server without removing config	No	false

Available MCP Tools

The Planning Analytics MCP server provides these tools

TM1 Cube Tools

- <https://www.ibm.com/docs/en/planning-analytics/3.1.0?topic=tools-tm1-cube>

TM1 Server tools

- <https://www.ibm.com/docs/en/planning-analytics/3.1.0?topic=tools-tm1-server>

Test the Connection

After restarting IBM Bob, verify the connection:

1. Open IBM Bob
2. Start a new conversation, make sure you set the
3. Ask: "What TM1 servers are available?"
4. IBM Bob should use the `get_available_tm1_servers` tool to query your PA instance

If successful, you'll see a response listing your TM1 servers.

Troubleshooting

è Just Ask Bob to fix it for you

Connection Issues

Problem: "Failed to connect to MCP server"

Solutions:

- Verify the URL is correct and accessible
- Check network connectivity (VPN if required)
- Ensure the server is running
- Verify firewall rules allow the connection

Authentication Issues

Problem: "401 Unauthorized" or "403 Forbidden"

Solutions:

- Verify credentials are correct
- Re-encode credentials in Base64
- Check that the Authorization header includes "Basic " prefix
- For SaaS, verify the API key is active and has proper permissions

Timeout Issues

Problem: Requests timing out

Solutions:

- Increase the timeout value (e.g., to 600 seconds)
- Check server performance and load
- Verify network latency

Tool Not Found

Problem: "Tool not available"

Solutions:

- Verify the PA server supports the MCP endpoint
- Check PA version compatibility
- Ensure the tool name is spelled correctly

Security Best Practices

1. **Never commit credentials** to version control
2. **Use environment-specific credentials** (dev, test, prod)
3. **Rotate API keys regularly** for SaaS deployments

4. **Use HTTPS** whenever possible
5. **Limit alwaysAllow tools** to only those you trust
6. **Set disabled: true** for servers not in use rather than deleting them

Advanced Configuration

Custom Timeout for Specific Servers

For servers with slower response times:

```
{
  "type": "streamable-http",
  "url": "http://slow-server.company.com/api/v0/agent-ai/cube/mcp",
  "timeout": 600,
  ...
}
```

Disabling a Server Temporarily

To temporarily disable without removing configuration:

```
{
  "my-pa-server": {
    "type": "streamable-http",
    "url": "...",
    "disabled": true
  }
}
```

Requiring Approval for All Tools

Remove the alwaysAllow array or set it to empty:

```
{
  "my-pa-server": {
    "type": "streamable-http",
    "url": "...",
    "alwaysAllow": []
  }
}
```

Support and Resources

- **IBM Planning Analytics Documentation:** [IBM Documentation](#)

- **MCP Protocol Specification:** [Model Context Protocol](#)
- **IBM Bob Support:** Contact your IBM representative

Appendix: Complete Configuration Template

```
{
  "mcpServers": {
    "YOUR_SERVER_NAME": {
      "type": "streamable-http",
      "url": "YOUR_PA_MCP_ENDPOINT",
      "headers": {
        "Authorization": "Basic YOUR_BASE64_CREDENTIALS"
      },
      "timeout": 300,
      "alwaysAllow": [
        "get_cubes_that_may_answer_query",
        "get_available_tm1_servers",
        "get_data_from_data_explorer",
        "lookup_potential_members",
        "get_cube_sample_members"
      ],
      "disabled": false
    }
  }
}
```

Replace:

- YOUR_SERVER_NAME: A descriptive name for your server
- YOUR_PA_MCP_ENDPOINT: Your PA MCP endpoint URL
- YOUR_BASE64_CREDENTIALS: Your Base64-encoded credentials

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